

## LAB EXPERIMENT

### SET 15: PRODUCT SALES ANALYSIS

|                                  |                                           |
|----------------------------------|-------------------------------------------|
| <b>Department</b>                | AIDS                                      |
| <b>Course Code / Course Name</b> | DSA0602 / Data Handling and Visualization |
| <b>Experiment</b>                | Data Visualization using R and Tableau    |
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| <b>Date</b>                      | 8 October 2026                            |
| <b>Day</b>                       | Thursday                                  |

### Aim

To analyze monthly product sales data using a grouped bar chart, a stacked area chart and a sales table in R, and to build an interactive dashboard using Tableau. Dataset: Monthly Product Sales

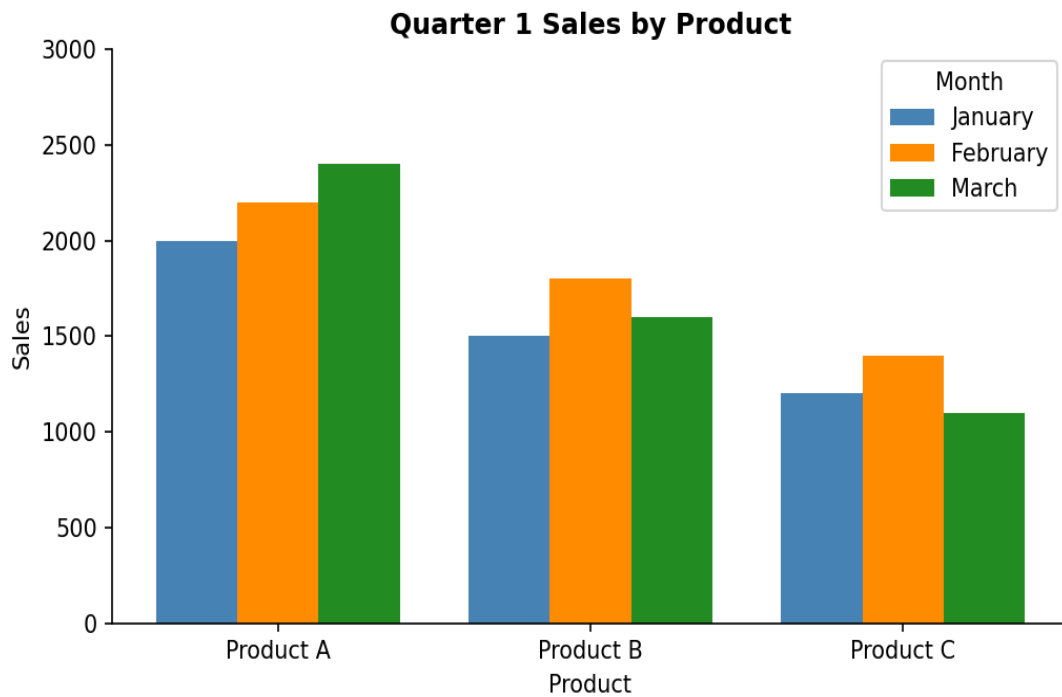
| Product ID | Product Name | January Sales | February Sales | March Sales |
|------------|--------------|---------------|----------------|-------------|
| 1          | Product A    | 2000          | 2200           | 2400        |
| 2          | Product B    | 1500          | 1800           | 1600        |
| 3          | Product C    | 1200          | 1400           | 1100        |

### PROGRAM 1: Grouped Bar Chart of Quarterly Sales

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

```
# Dataset (rows = months, columns = products)
sales_matrix <- rbind(
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)
colnames(sales_matrix) <- c("Product A", "Product B", "Product C")

# Grouped Bar Chart
barplot(sales_matrix,
  beside = TRUE,
  col = c("steelblue", "darkorange", "forestgreen"),
  xlab = "Product",
  ylab = "Sales",
  main = "Quarter 1 Sales by Product",
  legend.text = rownames(sales_matrix),
  args.legend = list(x = "topright", title = "Month"),
  ylim = c(0, 3000))
```

**OUTPUT:****Output 1: Quarter 1 Sales by Product (Grouped Bar Chart)****Observation:**

Product A has the highest sales in every month and grows steadily from 2,000 in January to 2,400 in March. Products B and C both rise in February but fall in March (B to 1,600 and C to 1,100). Product C has the lowest sales overall, with its March value of 1,100 being the lowest figure in the dataset.

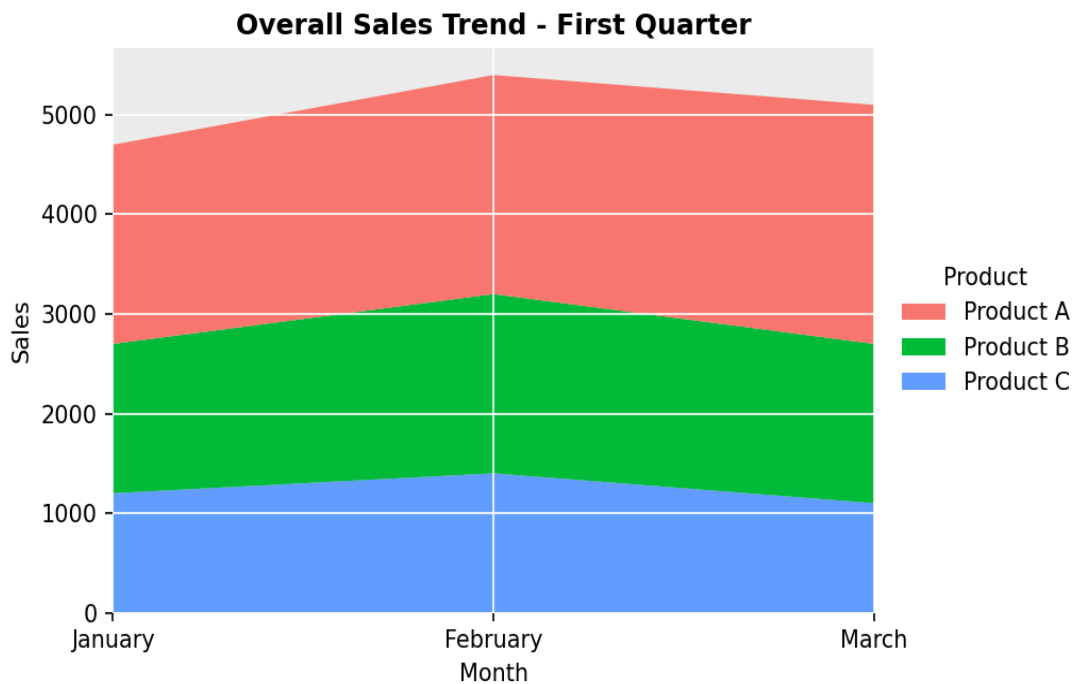
**PROGRAM 2: Stacked Area Chart of Overall Sales Trend**

Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
library(ggplot2)

# Dataset in long format
long_sales <- data.frame(
  month = rep(1:3, times = 3),
  product = rep(c("Product A", "Product B", "Product C"), each = 3),
  sales = c(2000, 2200, 2400,
            1500, 1800, 1600,
            1200, 1400, 1100)
)

# Stacked Area Chart
ggplot(long_sales, aes(x = month, y = sales, fill = product)) +
  geom_area() +
  scale_x_continuous(breaks = 1:3,
                    labels = c("January", "February", "March")) +
  labs(title = "Overall Sales Trend - First Quarter",
       x = "Month",
       y = "Sales",
       fill = "Product")
```

**OUTPUT:****Output 2: Overall Sales Trend (Stacked Area Chart)****Observation:**

The total sales of all products increase from 4,700 in January to 5,400 in February (about 15%) and then fall to 5,100 in March (about 6%). February is the best month overall. Product A forms the largest layer of the area, about 43% of the quarter's total sales of 15,200, followed by Product B (about 32%) and Product C (about 24%).

**PROGRAM 3: Table of Monthly Sales for Each Product**

Build a table to show the monthly sales data for each product. Label the table elements.

```
# Dataset
sales <- data.frame(
  Product_ID = 1:3,
  Product_Name = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)

# Add a total column for the quarter
sales$Total <- sales$January + sales$February + sales$March

# Display the table
print(sales)

# Monthly totals across all products
colSums(sales[, c("January", "February", "March", "Total")])
```

**OUTPUT:**

|   | Product_ID | Product_Name | January | February | March | Total |
|---|------------|--------------|---------|----------|-------|-------|
| 1 | 1          | Product A    | 2000    | 2200     | 2400  | 6600  |

|         |   |           |       |       |      |      |
|---------|---|-----------|-------|-------|------|------|
| 2       | 2 | Product B | 1500  | 1800  | 1600 | 4900 |
| 3       | 3 | Product C | 1200  | 1400  | 1100 | 3700 |
| January |   | February  | March | Total |      |      |
| 4700    |   | 5400      | 5100  | 15200 |      |      |

**Output 3a: Monthly Product Sales Table (R Console)**

| Product Name | January     | February    | March       | Total        |
|--------------|-------------|-------------|-------------|--------------|
| Product A    | 2000        | 2200        | 2400        | 6600         |
| Product B    | 1500        | 1800        | 1600        | 4900         |
| Product C    | 1200        | 1400        | 1100        | 3700         |
| <b>Total</b> | <b>4700</b> | <b>5400</b> | <b>5100</b> | <b>15200</b> |

**Output 3b: Monthly Sales Table with Totals****Observation:**

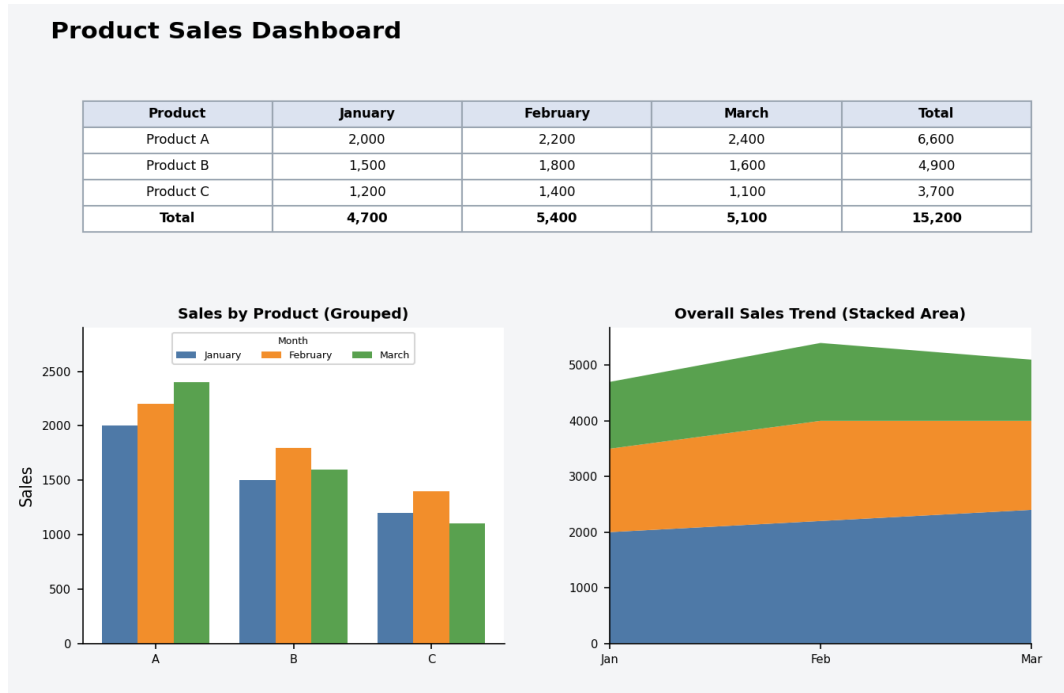
The table gives the exact sales of each product in each month. Product A has the highest quarterly total (6,600), then Product B (4,900) and Product C (3,700). The monthly totals show that February (5,400) was the strongest month and January (4,700) the weakest.

**PROGRAM 4: Interactive Product Sales Dashboard in Tableau**

Develop a Tableau dashboard combining the grouped bar chart, stacked area chart and the table for interactive exploration of sales data.

**Steps in Tableau**

1. Save the data as an Excel or CSV file and connect to it in Tableau (Connect > Microsoft Excel / Text file). Select the January Sales, February Sales and March Sales columns, right-click and choose Pivot, then rename the new fields to "Month" and "Sales".
2. Sheet 1 (Grouped Bar Chart): Drag Product Name and then Month to Columns, and Sales to Rows (SUM). Drag Month to Color and add Sales to Label. Rename the sheet "Sales by Product".
3. Sheet 2 (Stacked Area Chart): Drag Month to Columns and Sales to Rows (SUM). Change the Marks type to Area, drag Product Name to Color, and rename the sheet "Sales Trend".
4. Sheet 3 (Table): Drag Product Name to Rows, Month to Columns and Sales to Text. Use Analysis > Totals > Show Row Grand Totals and Show Column Grand Totals. Rename the sheet "Sales Table".
5. Create a new Dashboard (Dashboard > New Dashboard), set the size to Automatic, and drag the three sheets onto the dashboard area.
6. Add interactivity: select each chart and click Use as Filter, so clicking a product or month in one view filters the others. Add Product Name and Month filters (Show Filter > Apply to Worksheets > All Using This Data Source).
7. Add the title "Product Sales Dashboard" and test the filters in Presentation Mode.

**OUTPUT:****Output 4: Interactive Product Sales Dashboard (Tableau)****Observation:**

The dashboard shows the product comparison, the overall sales trend and the sales table together.

**Procedure**

1. Define the sales values for each product and month in R as a matrix and a data frame.
2. Use `barplot()` with `beside = TRUE` to create the grouped bar chart of first quarter sales.
3. Arrange the data in long format and use `ggplot2` with `geom_area()` to draw the stacked area chart.
4. Create a data frame of the monthly sales, add a Total column, and display it with `print()` and `colSums()`.
5. Load the sales file into Tableau, pivot the month columns, and build the grouped bar chart, stacked area chart and table worksheets.
6. Combine the worksheets in a Tableau dashboard and add filters and actions for interactivity.

**RESULT:**

The monthly product sales data was successfully analyzed using a grouped bar chart, a stacked area chart and a sales table in R. An interactive Tableau dashboard was also created, combining all three views so that users can explore the sales data by product and by month.

**Key Functions and Tools Used**

| Function / Tool                                  | Purpose                                       |
|--------------------------------------------------|-----------------------------------------------|
| <code>rbind()</code> / <code>data.frame()</code> | Creates the sales matrix and the sales table. |
| <code>barplot(beside = TRUE)</code>              | Creates the grouped bar chart.                |

| Function / Tool        | Purpose                                                 |
|------------------------|---------------------------------------------------------|
| ggplot() / geom_area() | Creates the stacked area chart.                         |
| print() / colSums()    | Displays the sales table and calculates monthly totals. |
| Tableau Pivot          | Converts the month columns into Month and Sales fields. |
| Tableau Dashboard      | Combines worksheets with filters and actions.           |